MINING DEALER BEST PRACTICE

Scavenge Pump Support Bracket for C-175 Engine

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management

November 2021
1.04-211125-1

1.0 Introduction

Mining equipment is part of the production process, playing a key role in achieving the production goals of any mining operation. To ensure the availability and reliability of the equipment, it is necessary to include maintenance and repair tasks, tasks that must be performed with efficiency, quality, and safety. In large mining equipment, such as the CAEX 797F, these tasks include, among others, the handling of large and heavy components.
To achieve the objectives of safety, efficiency and quality, the maintenance department must ensure that its personnel have the necessary training, support information (technical and procedural), infrastructure, tools, and support equipment.
This best practice was designed to improve the safety of the C-175 engine scavenge pump replacement task.

2.0 Best Practice Description
2.1 Opportunity Detection

During the analysis of the task of removing and installing the Scavenge pump on the C-175 engine, opportunities for improvement were detected. These opportunities motivated the development of the accessory proposed in this Best Practice.

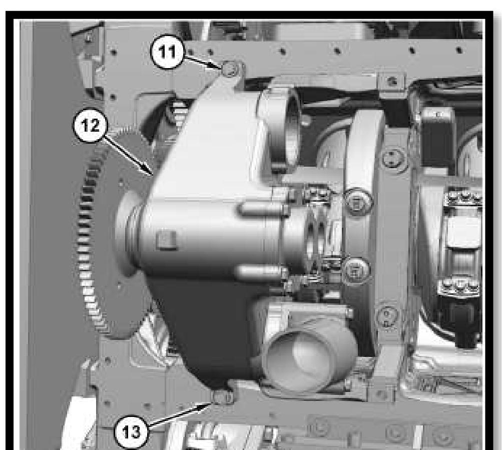
A support bracket for the Scavenge pump of the C-175 engine is manufactured, to provide greater safety for the task.

Areas of Opportunity

Safety

In SIS, it states:

Use a suitable lifting device to remove the Scavenge pump (12). The weight of the oil Scavenge pump (12) is approximately 80 kg (176 lb). Lift and position the Scavenge pump (12) to align it with the engine cylinder block. (KENR6052).
But it does not show or tell with which tool to remove it.



Time (Efficiency)

According to our records / repair history, the time this task (Scavenge Pump Replacement) requires, following the recommended procedure, on average 12 hours of downtime per machine.

Resources (Labor)

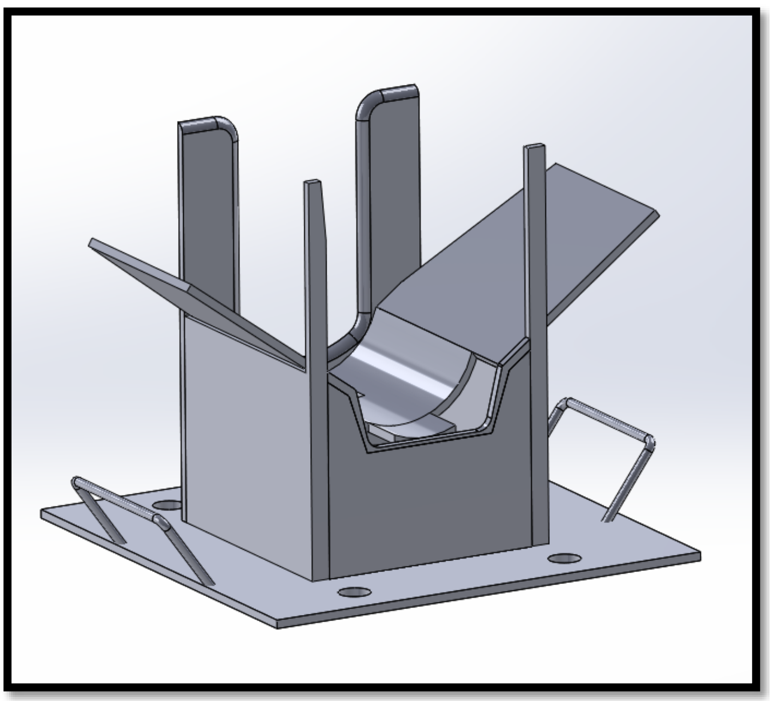
Normally 2 technicians are assigned to perform this task. This means that in total we use 24-man hours to complete the task, following the recommended procedure.

Quality

The observations made to the replacement of the Scavenge Pumps also detected some substandard events related to quality, which were mainly generated by the lack of protection of the equipment's components.

2.2 Proposed Improvement Solution

The proposed solution is the use of a Support, specially designed for this application (Figure 1). The function of this support is to offer a safe and adequate support to the Scavenge Pump and to avoid unexpected movements at the time of performing the task.



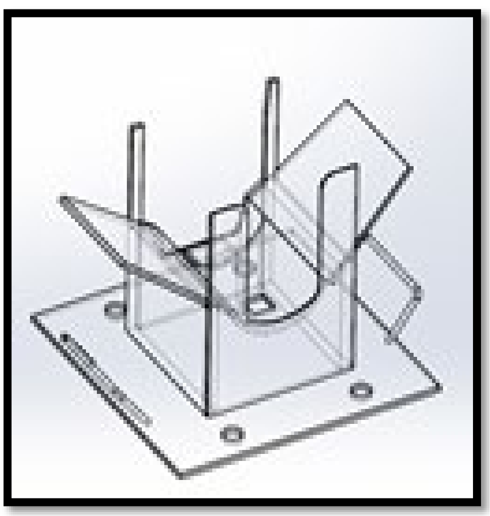
3.0 Implementation Steps

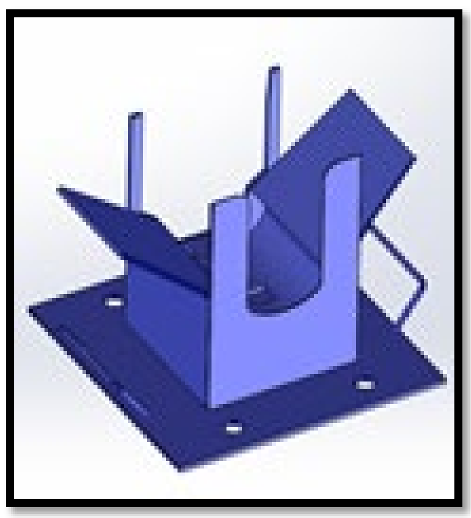
The general steps to be followed for the implementation of this Best Practice are as follows:

3.1 Fabrication of Scavenge Pump support bracket.
3.2 Technical certification of the support bracket.
3.3 Practical tests in tasks of Scavenge Pump replacement.
3.4 Final adjustments
3.5 Documentation of the SOP (Standard Operating Procedure).
3.6 Personnel training
3.7 Commissioning and Validation of Benefits

Design: The first stage is the design of the support bracket. This must be adapted to the contour of the pump, taking into consideration not to interfere with the rest of the components that surround the pump and adapting to the reduced space that exists in that sector. In addition, this support bracket of the Scavenge pump must be mounted on a head that has movement in the three axes and then mount the whole assembly (pump support bracket + rotating head) on a hydraulic platform to have up and down displacement.

<u>Design Stages:</u>





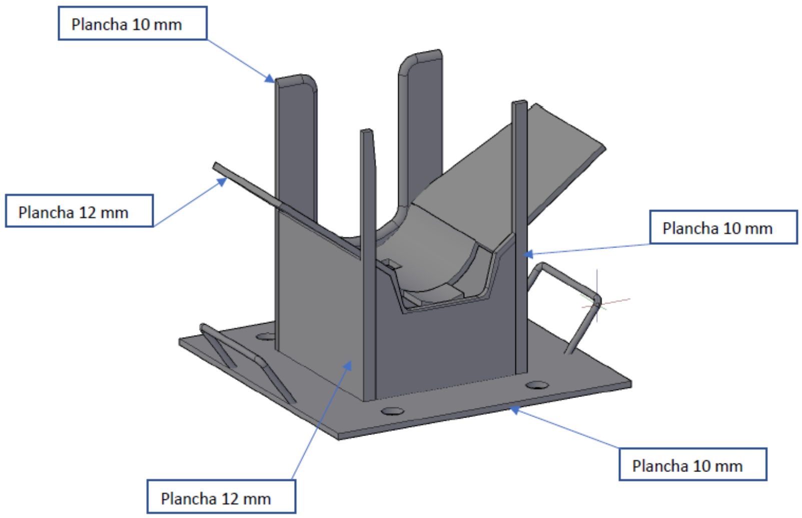
SolidWorks design

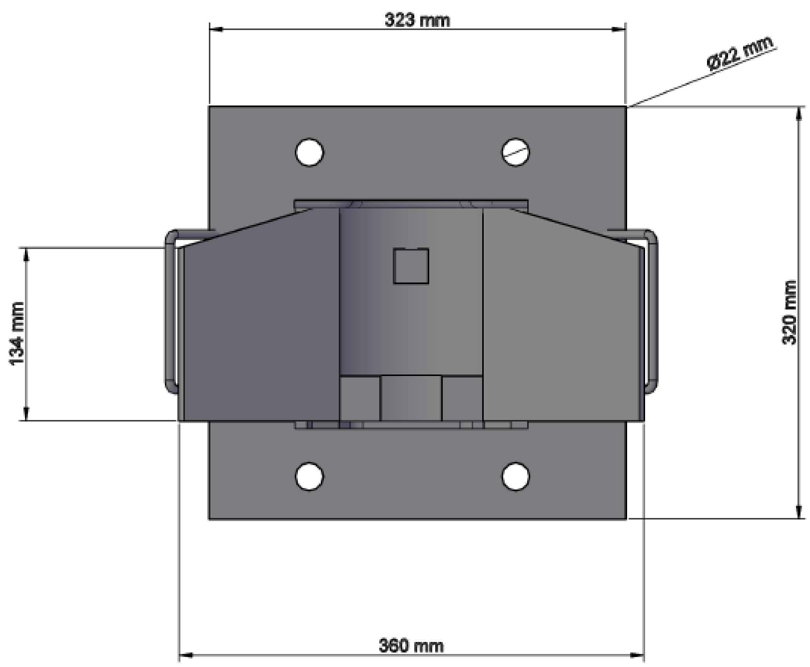
Actual Size

Support + head + platform assembly

3.1 Tool Manufacture " Scavenge Pump support bracket for C-175 Engine".

The Tool was manufactured locally according to the following design specifications. This design was performed locally in the Continuous Improvement Department of the Finning - DMH contract. It is considered the appropriate size to be able to hold the Scavenge Pump, without interfering with any point of the 797F structure.





Dimensioned drawings

Note: In our case, manufacturing was done with a local supplier.

RUT: 76.252.489-9
PIRITA 12450 GALPON 26, PARQUE CIE
TELEFONO: 55-2437454 – ANTOFAGASTA

3.2 Technical certification of the C-175 Motor Scavenge Pump support bracket.

It is important to certify the welds and capacity analysis of the accessory once it has been manufactured.

MC-GR-04619-HSBM

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REV	DESCRIPCION	INGENIERIA			APROBACION	
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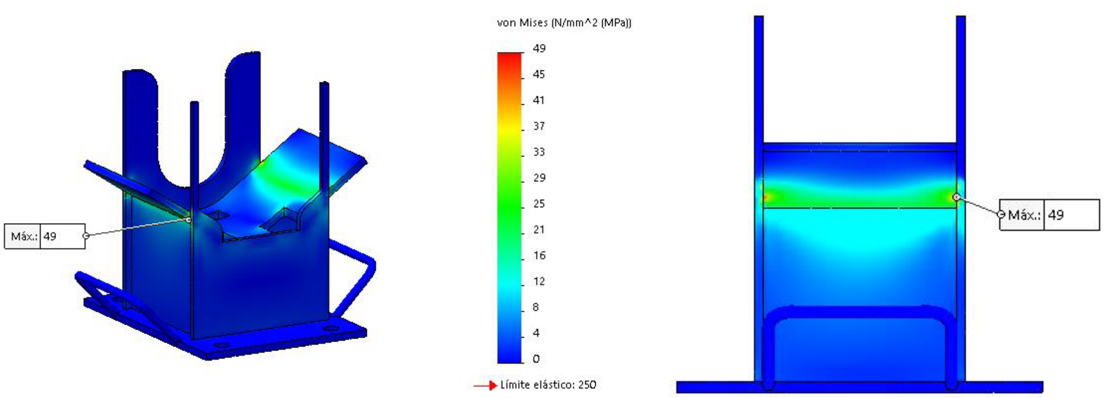
0	GR Ingeniería.	Giovani Rojas	Giovani Rojas	07-05-19		
A	Revisión Interna	Ricardo Vega V.	Ricardo Vega V.	07-05-19		

structural design analysis for
Scavenge Pump support
bracket

The calculation is developed by Finite Element Method, applying the mathematical model described in point 5, the boundary conditions indicated in point 6.1, on the discretized domain of point 6.2, as a result the following stress distribution in the element is obtained:

Maximum Stress = 49 MPa.
Maximum stress location = 10 mm plate.



3.3 Practical tests on Scavenge Pump change-over tasks.

Once the pump support bracket has been received, we recommend performing operation tests in the workshop, in a controlled and safe environment, to identify and make final adjustments if required.

3.4 Final adjustments.

It is important to introduce final adjustments after testing. In our case no deviations were detected.

3.5 SOP Documentation (Standard Operating Procedure)

To ensure proper operation of this accessory, we recommend documenting the standard operating procedure (SOP).
We also recommend reviewing the safety standards particular to your operation to adapt the SOP to your specific needs.

3.6 Staff Training

Adequate training of personnel who will routinely use the Tool is important for the Scavenge Pump replacement tasks. Consider at least 2 technicians per shift.
We recommend that this training include review of the standard operating procedure (SOP) as well as practice in a safe and controlled environment.

3.7 Start-up and Benefits Validation

Once the tool has been put into operation, the correct use and recording of benefits must be followed up. Once the expected benefits have been obtained, the Standard Jobs must be improved / modified with the Planning and Planning department, especially the new Machine Down Times and Man Hours.

4.0 Benefits

Based on our observations and records of the use of this accessory in our operation, the benefits documented so far reflect the following improvements (by equipment).

	Risk Level without Tool	Risk Level with Tool	Number of Technicians without Tool	Number of Technicians with Tool	Total machine downtime without tool	Total machine downtime with tool	Man hours without tool	Man hours with tool
Engine Scavenge Pump Support	High	**Low**	2	**2**	12	**10**	24	**20**

As can be seen in the table of results, the benefits are in the areas of Safety in the execution time and in the machine hours of the Task Execution.

Safety: The clamping maneuver is improved by developing a tight support for the Scavenge pump.

Labor time (Machine down hours): The reduction of the time of the task reaches 2 hours or 16%. It means that the task of replacing the pump is done in 10 hours, 83% of the time previously used.

It is important to highlight that the improvement in the task completion time has a direct impact on the MTTR result of the fleet. Its influence should be analyzed in conjunction with the frequency of Scavenge Pump replacement.

Man Hours (MH used): With relation to the number of Technicians used in the task, it is reduced by one.

Quality: Although it is difficult to quantify, we have detected improvements in the quality of the Scavenge Pump replacement operation. The control of the maneuvers and the effort of the personnel positively influence the quality of the work performed.
Improvements in this sense, Quality, will positively influence the MTBS results of the fleet.

5.0 Resources Required

The resources required are those specified in this Best Practice.
The manufacturing cost of the tool was $ 1,999,200. - CLP, approximately **USD$2,480.71.**

6.0 Supporting Attachments / References

The following information is attached to this Best Practice

- Schematic / Fabrication Drawing (see document appendix)

Note: Structural calculations and certification can be provided upon request.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

Best Practice Generation, Implementation and Documentation

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